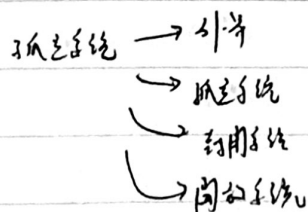


量子统计

2021. 11. 24

系统的宏观量是微观量的统计平均

热力学:



平衡态: 孤立系统经历长时间后, 宏观量不随时间变化的状态

状态参量: 确定平衡状态的一组互相独立宏观量

几何参量: V

力学: P

化学: μ, ν, N

电磁: $E, \vec{P}, \dots$

准静态过程: 过程进行缓慢, 每一个中间态都是平衡态

$$dE = dW + dQ + \sum_j \mu_j dN_j$$

热容量 $C = \frac{dQ}{dT}$

统计力学的 basic concept and 近独立粒子系统

微观量: 单个粒子性质的物理量

宏观量: 描述大量粒子组成的系统的宏观性质的量

力学规律 与 统计规律

正则运动 $q_i(t), p_i(t)$ 描述 $H(q_i, p_i)$

$\uparrow$
广义
坐标

$\uparrow$
广义
动量

$$\frac{dq_i}{dt} = \frac{\partial H}{\partial p_i}$$

$$\frac{dp_i}{dt} = -\frac{\partial H}{\partial q_i}$$

μ 空间: 由度类在相空间和自身在相空间的 $2r$ 维空间

$$\left(\begin{array}{l} \vec{q} \\ \vec{p} \end{array} \right) \leftarrow \begin{array}{l} \text{坐标} \\ \text{动量} \end{array}$$

相点: 粒子的一个运动状态

相轨道: 粒子运动状态的演化

相体积: 等能面附近的体积

$$\mu = \iiint_V dx dy dz \iiint_{H \leq E} dP_x dP_y dP_z$$

相体积包含微体积元 $\frac{\mu_0}{h_0^r} \leftarrow \text{微元}$

- 1D 谐振子

$$\gamma=1, \quad \varepsilon = \frac{p_x^2}{2m} + \frac{1}{2}m\omega^2 x^2 \Rightarrow \frac{p_x^2}{2m\varepsilon} + \frac{x^2}{\frac{2\varepsilon}{m\omega^2}} = 1$$

$$\mu = \pi \cdot \int_{-\infty}^{\infty} \sqrt{\frac{2\varepsilon}{m\omega^2}} = \frac{2\pi\varepsilon}{\omega}$$

转动, $\gamma=2, \quad \theta, \varphi, p_\theta, p_\varphi$

$$p_\theta = mr^2 \dot{\theta}$$

$$\varepsilon = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2)$$

$$p_\varphi = mr^2 \sin^2\theta \dot{\varphi}$$

$$= \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\dot{\varphi}^2)$$

$$= \frac{1}{2I} (p_\theta^2 + \frac{p_\varphi^2}{\sin^2\theta})$$

$$\int_{\varepsilon} dp_\theta dp_\varphi = 2\pi I \varepsilon \sin\theta$$

$$\begin{aligned} \mu &= \iiint d\theta d\varphi dp_\theta dp_\varphi \\ &= \int_0^{2\pi} d\varphi \int_0^\pi d\theta \cdot 2\pi I \varepsilon \sin\theta \\ &= 4\pi I \varepsilon \cdot 2\pi = 8\pi^2 I \varepsilon \end{aligned}$$

(二) 量子力学描述 $\hat{H}(\hat{q}, \hat{p})$

(三) 统计力学描述: 在一定的宏观条件下, 某一时刻系统以概率分布的形式处于各种可能的微观状态中。在平衡态下, 概率分布不变

宏观量的统计性质: (微观量) 的统计平均

数量 Δn 是有限值。 ΔV 是宏观量级, Δt 是宏观时间

统计力学不对时间做平均, 而是对一切可能状态做平均

理想内能是微观系统能量平均值

氣體压力是冲量平均值

(1) 没有定, 微分, 如温度, 压强

(2) 温度是外参量

近独立粒子系统: 粒子间的相互作用平均能量远小于平均动能

Ex: 理想氣體

统计平衡是由相互作用所致

不考虑粒子结构 (分子, 原子, 光子, 自由电子 —)

单粒子 (电子, 质子, 核子, 强子 —)

$$E = \sum_i n_i \varepsilon_i \quad \left[\begin{array}{l} \text{粒子能量} \\ \text{粒子在能级的分布情况 (number)} \end{array} \right]$$

Boltzmann 分布

Bose-Einstein 分布

Fermi-Dirac 分布



量子统计

2021.11.21

量子力学讲义, 统计力学

分布与微观状态: ϵ_i 在 n_i 个, ϵ_i 在 n_i 内能级

一个分布能多到什么程度

Boltzmann 系统 (粒子可区分) (粒子可分别)

定域系统 $W_{M,D} \{n_i\} = \frac{N!}{\prod_i n_i!} \prod_i g_i^{n_i}$

Bose = Einstein 分布子

Fermi-Dirac 分布

$$W_i = \binom{n_i + g_i - 1}{g_i - 1}$$

$$W_i = \binom{g_i}{n_i} \quad n_i \leq g_i$$

所有都是以“宏观量是微观量的统计平均”为出发点

Bose 分布

求分布出现的概率

“对于处于平衡态的孤立系统, 各个微观态出现的概率相等” Boltzmann
等概率假设

$P\{n_i\} = \text{常数} \times W\{n_i\} \leftarrow$ 热力学概率

最可几方法 $\bar{u} = \frac{\sum_i u\{n_i\} P\{n_i\}}{\sum_i W\{n_i\}} = \frac{\sum_i u\{n_i\} W\{n_i\}}{\sum_i W\{n_i\}}$

在 $W\{n_i\}$ 中有一个很高峰, 则称最可几分布。



Bose 分佈 $W_B\{n_i\} = \prod_i \binom{n_i + g_i - 1}{g_i - 1}$

$$\ln W_B\{n_i\} = \sum_i \ln \binom{n_i + g_i - 1}{g_i - 1} = \sum_i [\ln(n_i + g_i - 1)! - \ln(g_i - 1)! - \ln(n_i!)]$$

設 $g_i \gg 1$, $n_i \gg 1$, 利用 Stirling 公式: $\ln M! \approx M \ln M - M$

$$\begin{aligned} \ln W_B\{n_i\} &\approx \sum_i \left[(n_i + g_i - 1) \ln(n_i + g_i - 1) - (g_i - 1) \ln(g_i - 1) - n_i \ln(n_i) \right. \\ &\quad \left. - (n_i + g_i - 1) + (g_i - 1) + n_i \right] \\ &= \sum_i \left[n_i \ln \left(1 + \frac{g_i - 1}{n_i} \right) + (g_i - 1) \ln \left(1 + \frac{n_i}{g_i - 1} \right) \right] \end{aligned}$$

利用 Lagrange 乘数法

$$\begin{aligned} F(W_B\{n_i\}, \alpha, \beta) &= \sum_i \left[n_i \ln \left(1 + \frac{g_i}{n_i} \right) + g_i \ln \left(1 + \frac{n_i}{g_i} \right) \right] + \alpha \left(N - \sum_i n_i \right) \\ &\quad + \beta \left(E - \sum_i n_i \epsilon_i \right) \end{aligned}$$

$$\frac{\partial F}{\partial n_i} = \ln \left(1 + \frac{g_i}{n_i} \right)$$

$$\ln(n_i + g_i - 1) + 1 - \ln(n_i) - 1 - \alpha - \beta \epsilon_i = 0$$

$$\ln \left(1 + \frac{g_i - 1}{n_i} \right) = \alpha + \beta \epsilon_i$$

$$1 + \frac{g_i - 1}{n_i} = \exp(\alpha + \beta \epsilon_i)$$

$$\Rightarrow n_i = \frac{g_i - 1}{\exp(\alpha + \beta \epsilon_i) - 1} \approx \frac{g_i}{\exp(\alpha + \beta \epsilon_i) - 1}$$

$$= g_i \frac{1}{\exp(\alpha + \beta \epsilon_i) - 1}$$

↑
能级占据概率



量子统计

2021. 12. 01

半经典分布 $n_i = \frac{g_i}{e^{\alpha + \beta \epsilon_i} \mp 1}$ (- : Bose, + : Fermi)

$$\approx g_i e^{-(\alpha + \beta \epsilon_i)} \quad e^{\alpha} \gg 1 \Rightarrow n_i \ll g_i$$

$$\begin{aligned} \ln W_F\{n_i\} &= \sum_i \ln \frac{g_i!}{n_i! (g_i - n_i)!} \\ &= \sum_i \left[g_i (\ln g_i - 1) - n_i (\ln n_i - 1) - (g_i - n_i) (\ln (g_i - n_i) - 1) \right] \\ &= \sum_i \left[g_i \ln \frac{g_i}{g_i - n_i} - n_i \ln \frac{n_i}{g_i - n_i} \right] \\ &= \sum_i \left[n_i \ln \left(\frac{g_i}{n_i} - 1 \right) - g_i \ln \left(1 - \frac{n_i}{g_i} \right) \right] \\ &\approx \sum_i n_i \ln \frac{g_i}{n_i} + n_i \end{aligned}$$

$$\begin{aligned} \ln W_S\{n_i\} &= \sum_i n_i \left(\ln \frac{g_i}{n_i} + 1 \right) = \sum_i \left[n_i \ln g_i - n_i (\ln n_i - 1) \right] \\ W_S\{n_i\} &\approx \frac{1}{N!} \prod_i \frac{g_i^{n_i}}{n_i!} \end{aligned}$$

定域粒子：兩個粒子的局限範圍不重疊，則可認為兩粒子可區分

Boltzmann 分布： $W_{\text{Boltzmann}}\{n_i\} = N! \prod_i \frac{g_i^{n_i}}{n_i!}$

$$\begin{aligned} \ln W_{\text{Boltzmann}}\{n_i\} &= \ln N! + \sum_i n_i \ln g_i - n_i (\ln n_i - 1) \\ \frac{\partial \ln W_{\text{Boltzmann}}\{n_i\}}{\partial n_i} &= + \ln g_i - \ln n_i + 1 = \ln \frac{g_i}{n_i} \end{aligned}$$

$$\ln \frac{g_i}{n_i} - \alpha - \beta \epsilon_i = 0 \Rightarrow n_i = g_i e^{-\alpha - \beta \epsilon_i}$$



α 和 β 的物理意义

令 $\{n_i\}$ 和 $\{n_i''\}$ 的两个近独立系统 1 和 2 进行热交换 (能级不变)

形成新系统. 能级 1 的分布是 $\{\bar{n}_i'\}$, 另一为 $\{\bar{n}_i''\}$

$$N' = \sum_i \bar{n}_i' = \sum_i n_i'$$

$$N'' = \sum_i \bar{n}_i'' = \sum_i n_i''$$

$$E = \sum_i \bar{n}_i' \epsilon_i' + \sum_j \bar{n}_j'' \epsilon_j''$$

$$W_{12} = \prod_i \frac{g_i'^{\bar{n}_i'}}{\bar{n}_i'!} \prod_j \frac{g_j''^{\bar{n}_j''}}{\bar{n}_j''!}$$

$$\ln W_{12} = \sum_i [\bar{n}_i' \ln g_i' - \bar{n}_i' (\ln \bar{n}_i' - 1)] + \sum_j [\bar{n}_j'' \ln g_j'' - \bar{n}_j'' (\ln \bar{n}_j'' - 1)]$$

$$\frac{\partial W_{12}}{\partial \bar{n}_i'} = \ln g_i' - \ln \bar{n}_i' = \ln \frac{g_i'}{\bar{n}_i'}$$

$$\frac{\partial W_{12}}{\partial \bar{n}_j''} = \ln \frac{g_j''}{\bar{n}_j''}$$

$$\Rightarrow \bar{n}_i' = g_i' e^{-\alpha' - \beta \epsilon_i'} \\ \bar{n}_j'' = g_j'' e^{-\alpha'' - \beta \epsilon_j''}$$



量子统计

2021. 11. 26

2. (6.1) $\hat{H} = \frac{\hat{p}^2}{2m}$

$$\mu = \iiint_V dx dy dz \iiint_{H \leq \epsilon} dp_x dp_y dp_z$$
$$= V \cdot \frac{4}{3} \pi (2m\epsilon)^{\frac{3}{2}}$$

$$D(\epsilon) = \frac{d\mu}{h^3 d\epsilon} = V \cdot \frac{4}{3} \pi \cdot (2m)^{\frac{3}{2}} \cdot \frac{3}{2} \epsilon^{\frac{1}{2}} / h^3 = \frac{2\pi V}{h^3} (2m)^{\frac{3}{2}} \epsilon^{\frac{1}{2}}$$

3. (6.2) $\hat{H} = \frac{\hat{p}_x^2}{2m}$

$$\mu = \int_L dx \int_{H \leq \epsilon} dp_x = L \cdot 2 \times (2m\epsilon)^{\frac{1}{2}}$$

$$D(\epsilon) = \frac{d\mu}{h d\epsilon} = 2L \times (2m)^{\frac{1}{2}} \cdot \frac{1}{2} \epsilon^{-\frac{1}{2}} / h = \frac{2L}{h} \left(\frac{m}{2\epsilon} \right)^{\frac{1}{2}}$$

4. (6.3) $\hat{H} = \frac{\hat{p}^2}{2m}$

$$\mu = \iint_{L \times L} dx dy \iint_{H \leq \epsilon} dp_x dp_y = L^2 \cdot \pi \cdot 2m\epsilon$$

$$D(\epsilon) = \frac{d\mu}{h^2 d\epsilon} = \frac{2\pi m}{h^2} L^2$$

5. (6.4) $\hat{H}^2 = c^2 \hat{p}^2$

$$\mu = \iiint_V dx dy dz \iiint_{H \leq \epsilon} dp_x dp_y dp_z$$

$$= V \cdot \frac{4}{3} \pi \left(\frac{\epsilon}{c} \right)^3$$

$$D(\epsilon) = \frac{d\mu}{h^3 d\epsilon} = \frac{4\pi V}{(ch)^3} \epsilon^2$$



6. (補充題 1) ~~Why $W(n) = \frac{N!}{n!}$~~

$$W(n) = \binom{N}{n} = \frac{N!}{n!(N-n)!}$$

$$\Delta S = k_B \ln W(n) = k_B (\ln N! - \ln n! - \ln (N-n)!) \quad \text{}$$

$$\Delta E = nW$$

在熱平衡條件下, $\Delta F = \Delta E - T\Delta S - S\Delta T = -P\Delta V \Rightarrow \frac{\partial \Delta F}{\partial n} = 0$

$$\Rightarrow \Delta F = nW - T k_B \ln \frac{N!}{n!(N-n)!}$$

$$\approx nW - k_B T (N \ln N - n \ln n - (N-n) \ln (N-n))$$

$$\frac{\partial \Delta F}{\partial n} = W + k_B T (\ln n - \ln (N-n)) = 0$$

$$\frac{W}{k_B T} = \ln \left(\frac{N}{n} - 1 \right) \Leftrightarrow n = \frac{N}{1 + \exp\left(\frac{W}{k_B T}\right)} \approx N \exp\left(-\frac{W}{k_B T}\right)$$

7. (補充題 2) $W(n) = \binom{N}{n} \binom{N'}{n} = \frac{N!}{n!(N-n)!} \frac{N'!}{n!(N'-n)!}$

$$\Delta S = k_B \ln W(n)$$

$$\approx k_B [N \ln N - n \ln n - (N-n) \ln (N-n) + N' \ln N' - (N'-n) \ln (N'-n)]$$

$$\Delta E = nW = n(\epsilon_2 - \epsilon_1)$$

$$\Delta F = nW - k_B T [N \ln N + N' \ln N' - 2n \ln n - (N-n) \ln (N-n) - (N'-n) \ln (N'-n)]$$

$$\frac{\partial \Delta F}{\partial n} = W + k_B T [2 \ln n - \ln (N-n) - \ln (N'-n)] = 0$$

$$\frac{W}{k_B T} = \ln \left(\frac{N-n}{n} \right) + \ln \left(\frac{N'-n}{n} \right),$$

$$n^2 \exp\left(\frac{W}{k_B T}\right) = n^2 - (N+N')n + NN'$$

$$n = \frac{-(N+N') \pm \sqrt{(N+N')^2 + 4 \left(\exp\left(\frac{W}{k_B T}\right) - 1 \right) NN'}}{2 \left(\exp\left(\frac{W}{k_B T}\right) + 1 \right)}$$

若 $W \gg k_B T$, 則 $n \approx \sqrt{NN'} \exp\left(-\frac{W}{2k_B T}\right) = \sqrt{NN'} \exp\left(-\frac{\epsilon_2 - \epsilon_1}{2k_B T}\right)$